

# Alisa Sheinkman

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## Profile

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Researcher specializing in Bayesian machine learning, scalable inference algorithms and robust modelling frameworks. My work focuses on improving the reliability, calibration, interpretability and uncertainty quantification. Experienced in developing complex Bayesian models and novel inference algorithms.

## Experience

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**Research Scientist, 40folds**, Paris, France 2026 - Present

- Developed novel Bayesian Media Mix Models and time-series forecasting algorithms, translating probabilistic outputs into actionable business strategies.
- Designed robust validation frameworks, advanced model averaging, and ensembling techniques.
- Wrote scalable, production-ready code using Python, JAX and PyTorch.

**Doctoral Researcher in ML, University of Edinburgh**, Edinburgh, UK 2020 - 2025

**Thesis:** *Probabilistic Inference in Bayesian Neural Networks*

**Supervisor:** Dr. Sara Wade

- Developed a **novel class of Bayesian Neural Networks** (BNNs) that unifies model selection and inference, addressing the trade-off between accuracy and uncertainty quantification.
- Derived a scalable **Variational Inference** algorithm for this architecture, enabling efficient approximation of complex posteriors.
- Established new benchmarks for BNN architectural choices, demonstrating that Bayesian model combination significantly enhances calibration and reliability in safety-critical applications.

**Graduate Teaching Assistant, King's College London**, London, UK 2022 - 2025

Led tutorials and marked assessments for Probability and Statistics courses.

**Data Analyst, Nielsen IQ**, Moscow, Russia 2020 - 2021

Built data pipelines to analyze retail measurement data, extracting actionable market trends.

**Research intern, NRU HSE**, Moscow, Russia 2018 - 2019

Developed advances on braid groups and quantum matrix algebras.

## Education

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**University of Edinburgh**, Edinburgh, UK 2020 - 2025

**Ph.D. in Statistics**

**University of Cambridge**, Cambridge, UK 2019 - 2020

**MASt in Mathematical Statistics (Merit)**

**National Research University Higher School of Economics**, Moscow, Russia 2015 - 2019

**Bachelor of Mathematics (First Class Honours)**

## Skills

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**ML/DL Frameworks** PyTorch, JAX, NumPyro, Transformers, LangGraph

**Data Science** NumPy, SciPy, pandas, scikit-learn, HuggingFace

**Tools** Git, GitHub, Kubernetes, LaTeX

**Languages** English (C2, full proficiency), Russian (native), French (B1)

## Publications

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1. E. Semenova, **A.Sheinkman**, T. J. Hitge, S.M.Hall, J.Cockayne, *SMRS: advocating a unified reporting standard for surrogate models in the artificial intelligence era*, **NeurIPS**, 2025.
2. **A.Sheinkman**, S.Wade, *Understanding the Trade-offs in Accuracy and Uncertainty Quantification: Architecture and Inference Choices in Bayesian Neural Networks*, in Proceedings of the **ECML PKDD**, 2025.
3. **A.Sheinkman**, S.Wade, *Variational Bayesian Bow tie Neural Networks with Shrinkage*, arXiv preprint, arXiv:2411.11132, 2024.
4. A.Mitchell, E.Brown, R.Deo, Y.Hou, J.Kirton-Wingate, J.Liang, **A.Sheinkman**, C.Soelistyo, H.Sood, K.Xing, A.Wongprommoon, W.Yip, F.Aletta, *Deep learning techniques for noise annoyance detection: Results from an intensive workshop at the Alan Turing Institute*, the Journal of the Acoustical Society of America 153, A262-A262, 2023.

## Talks

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ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases), Porto, Portugal, 2025.

The first workshop on Advances in post-Bayesian methods, London, UK, 2025.

CMStatistics (Conference on Computational and Methodological Statistics), London, UK, 2024.

ISBA (International Society for Bayesian Analysis) world meeting, Venice, Italy 2024.

BAYSM (Bayesian Young Statisticians Meeting), Venice, Italy, 2024.

Early Career Researches Day of the Centre for Statistics Annual Conference at the University of Edinburgh, Edinburgh, UK, 2024.

## Scholarships and grants

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EurIPS travel support grant, 2025.

Travel support grant for the First workshop on Advances of post-Bayesian methods, €250, 2025.

ProbAI Hub Doctoral Prize Fellowship finalist, not awarded, 2025.

Dr Laura Wisewell Fund scholarship, €1,000, 2024.

ISBA (International Society for Bayesian Analysis) junior travel award, 400\$, 2024.

LMS (London Mathematical Society) early career research grant, €450, 2024.

## Miscellaneous

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Publications and a full list of research activities are available on the [website](#).

OVD-info volunteer (human rights defence and media group).

Sports: boxing (2021 - now), professional figure skating (competed 2002 - 2014).